EXECUTION PLAN GUIDE Catalyst Optimizer Output

How to Read & Analyze PySpark Physical Plans

A Step-by-Step Decoder for Tree Syntax, Stage Boundaries, Operator Identifiers, and Shuffle Warning Signs

Tree Syntax & Flow WSCG Stages Shuffle Red Flags

CORE CONCEPT #1

Reading Order & Tree Syntax

↑ Bottom-to-Top Execution Flow

Physical plans read from bottom to top. Data ingestion occurs at the lowest leaf nodes, flowing upward into transformations and final output projections.

+-

Right Branch / Single Child

Denotes the final branch or single downstream stream.

:-

Left / Intermediate Child

Denotes a parallel incoming data branch in multi-input joins.

PLAN NAVIGATION

Decoding Tree Branches & Execution Topology

```
+- SortMergeJoin [key#74L], [key#78L]
   :- Sort [key#74L ASC NULLS FIRST]      ⟵ Left Stream (:- )
   :  +- Exchange hashpartitioning
   :     +- Filter isnotnull(key#74L)
   :        +- Scan LeftTable
   +- Sort [key#78L ASC NULLS FIRST]      ⟵ Right Stream (+- )
      +- Exchange hashpartitioning
         +- Filter isnotnull(key#78L)
            +- Scan RightTable
```

1. Multi-Input Branch Alignment

The colon alignment (`:`) defines sub-tree depth. Left streams execute under `:-` while right streams close with `+-`.

2. Identifying Exchange Shuffles

Check whether shuffles exist on both streams. Dual `Exchange` nodes mean full network redistribution on both sides.

3. Predicate Pushdown Verification

Ensure `Filter` operators drop nulls/unneeded rows directly above `Scan` before data hits the shuffle boundary.

DEEP DIVE

Operator Syntax & Internal Expression IDs

`key#74L` Expression ID & Type

L signifies Long datatype (64-bit int).

`ENSURE_REQUIREMENTS` Planner Constraints

Indicates Catalyst injected the shuffle operator automatically to fulfill partitioning requirements of downstream joins.

`HashAggregate` Partial vs Final

`partial_avg` runs locally before shuffle; final aggregation evaluates after network exchange.